Relational Actualism Gravity and Cosmology (RAGC):

Covariance, The Local Ledger, Empirical Predictions, and Mathematical Derivations in Actualism Gravity

Joshua F. Sandeman

Independent Researcher, Salem, Oregon

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Abstract

This paper formalizes the extension of Relational Actualism (RA) into the non-inertial and curved spacetime regime (RAGC). We propose that the spacetime manifold acts as an active, bandwidth-limited causal graph that drives the objective pruning of the quantum state space. Modelling spacetime as a Directed Acyclic Graph (DAG), we reconceive energy-momentum conservation as a strict *local ledger* balance condition at every actualization vertex. The paper makes three empirical predictions that are testable independently of the open mathematical derivations: (P1) the inferred value of H_0 should correlate with mean actualization density along the line of sight, explaining the Hubble Tension without dark energy; (P2) the vacuum energy contribution to the metric source term is structurally zero, not fine-tuned, with implications for CMB power spectra distinguishable from ΛCDM; and (P3) daughter spacetimes nucleated at causal severance boundaries inherit an effective cosmological constant determined by the local actualization flux at nucleation—a dynamical alternative to the anthropic landscape. Key mathematical results include: a spectral definition of the Actualization Projector ($\mathscr{P}_{\text{act}}$) with a proof of vacuum energy suppression (Theorem 8.4); and the Ledger Partition Theorem (Theorem 8.9), establishing independent conservation across causal severances, which has been **formally verified** in Lean 4/Mathlib.[1] Furthermore, the framework quantitatively prohibits causally silent dark matter (WIMPs, axions, sterile neutrinos): a 100 GeV WIMP generates 10^{25}–10^{38} times less actualization flux than equivalent baryonic matter, falling orders of magnitude below the threshold for measurable gravitational lensing. The conjectural mechanisms for flat galactic rotation curves and the Bullet Cluster offset—which follow from the same actualization-density field—are developed in the companion paper RADM [Sandeman, 2026f]. A new RA-native derivation of the causal set generative measure is presented: the Benincasa-Dowker sign criterion ($S_{\text{BDG}}(v) > 0$) as the discrete actualization filter yields a Poisson-CSG formula $c_k^{\text{RA}}(x) \propto e^{-\mu}\mu^k$ with $\mu = \lambda\, V_{\text{coh}}\, p_{\text{th}}$ that is local and environment-dependent. A bandwidth-partition model predicts $H_{\text{local}} \approx 73.1\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ from the KBC void and a new falsifiable consequence: the Eridanus supervoid ($\delta \approx -0.30$) should give $H \approx 75.9\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$. The Jacobson coefficient $\alpha = 2\pi$ is confirmed to within ∼2% when the full baryonic disk mass [Bland-Hawthorn & Gerhard, 2016] is used — consistent with $\alpha = 2\pi$ exactly within the systematic uncertainty on v_{flat} and r_d. Four mathematical derivations remain open and are precisely scoped for collaborators.

[1]The RA Lean 4 proof files (`RA_AQFT_Proofs_v10.lean` and `RA_Proofs_Lean4.lean`) will be made publicly available on arXiv submission. The underlying Lean-QuantumInfo library (Meiburg, Lessa, and Soldati, 2025; arXiv:2510.08672) is publicly available at `https://github.com/Timeroot/Lean-QuantumInfo`.

Keywords: Causal Set Theory · Actualization Projector · Local Ledger · Vacuum Energy · Hubble Tension · Graph Cut Theorem · Cosmological Constant · Dark Matter · Lean 4 Verification · Benincasa-Dowker Action · Poisson-CSG · Eridanus Supervoid

1 Introduction: The Limits of the Timeless Manifold

The reconciliation of General Relativity and Quantum Mechanics remains the central crisis of modern physics. However, this impasse may not merely be a failure of specific models, but a failure of the underlying mathematical language. Since the introduction of the Minkowski spacetime continuum, physics has relied on a formalism that spatializes time. By treating time as a reversible parameter t, standard physics enforces a "block universe" ontology—a timeless container in which the future mathematically exists simultaneously with the past.

Crucially, standard Quantum Field Theory (QFT) suffers from this same mathematical Platonism. Because QFT lacks a formal mathematical mechanism to represent the mere fact that things happen, it fails to distinguish between mathematical potentia and physical reality, granting full gravitational weight to unactualized virtual states. The true crisis in physics is not merely that GR and QM are incompatible; it is that both frameworks lack a mechanism for genuine becoming.

Relational Actualism (RA) proposes that time and causality are ontologically primitive. Building upon relational frameworks [Rovelli, 1996], actualization is not a subjective epistemic update; it is an objective, irreversible physical interaction derived directly from on-shell boson exchanges and causal severances [Sandeman, 2026a]. Spacetime is not a passive background; it is an active, relentlessly growing causal graph.

Time as the growing graph. A foundational ontological commitment of RAGC, distinct from its technical content and worth stating explicitly before the formalism begins, is this: *time is not a dimension of a background manifold through which events are located. Time is the process of the causal graph growing.*

In the block universe picture that special relativity appears to enforce, past, present, and future co-exist as equally real coordinate slices. The "flow" of time is then either illusory or a feature of consciousness rather than of physics. RA rejects this picture not on philosophical grounds but on ontological ones: if actualization events are genuinely irreversible — if the vertices written into the causal DAG cannot be unwritten — then the past is fixed, the future is genuinely open, and the present is the growing frontier at which possibilities become actualities. Temporal becoming is not a psychological phenomenon; it is the physical process of the graph growing.

This identification has three immediate consequences. First, the arrow of time requires no separate explanation. It is not a thermodynamic emergent phenomenon explained by low-entropy initial conditions; it is encoded in the framework's most primitive commitment, since actualization is by definition irreversible. Second, the distinction between time and causality dissolves: to say that A caused B is to say that A's actualization was a causal precursor to B's in the DAG — there is no additional fact about "temporal priority" beyond causal precedence. Third, the proper time of an observer is the count of actualization events along their worldline, not a coordinate parameter. Time dilation — both kinematic and gravitational — is the statement that different worldlines accumulate different numbers of actualization events per coordinate second (Section 4).

This paper proceeds as follows. Section 3 states the postulates of RAGC. Section 2 previews the full generative algorithm. Section 4 derives relativistic time dilation in RA terms. Section 5 develops the local ledger formulation of conservation. Section 6 analyzes causal severances and their cosmological implications. Section 7 states the empirical predictions of RAGC explicitly, separated from the open derivation tasks. Section 8 presents the mathematical derivations, each developed as far as currently tractable with hard walls identified precisely.

Note on formal verification. The Ledger Partition Theorem (Theorem 8.9) and the Markov Blanket Shielding Theorem have been formally verified in Lean 4 with Mathlib. The verification establishes that the discrete graph-theoretic core of the Graph Cut proof is logically watertight given the stated axioms. The Lean formalization is **completely free of** `sorry` **tags** for the Graph Cut Theorem and its supporting lemmas: the two sum-decomposition lemmas (`sum_outgoing_decompose`

and `sum_incoming_decompose`) have been fully proved using `Finset` membership arguments and the `arrow_of_time` condition.

A second Lean 4 file, `RA_AQFT_Proofs_v10.lean`, formalises the finite-dimensional AQFT proof program. It defines density matrices, unitary conjugation, and the quantum relative entropy via the continuous functional calculus (CFC), and proves `frame_independence` and `rindler_stationarity` as compiled theorems. The proof depends on three axioms: `vacuum_lorentz_invariant` (the Poincaré-invariance of the Minkowski vacuum, a QFT physics input), `Matrix.cfc_conj_unitary` (the CFC unitary-conjugation lemma, sourced from the Lean-QuantumInfo library [Meiburg et al., 2025]), and `petz_monotonicity` (the data processing inequality, proved in Lean-QuantumInfo). All source files are available from the author.

2 The Engine of Becoming: A Preview of the Generative Algorithm

The framework developed in this paper — the Local Ledger Condition, the Actualization Projector, the metric sourcing equation, the time dilation account — all follow from a single underlying algorithm that has not yet been stated in its most complete form. The full treatment is the subject of the companion paper *The Engine of Becoming* [Sandeman, 2026e], but a preview is essential here because it reveals that RAGC's individual results are not separate claims but consequences of one unified generative process.

The algorithm is a **Stochastic Graph Rewriting System**. It replaces both the Schrödinger equation and the Einstein field equations with a single iterative procedure operating on the causal DAG. It is the precise formal answer to the question: what is the universe *doing*, step by step, as the causal graph grows?

The five steps. Let the universe at step n be defined by the graph $G_n = (V_n, E_n)$, where V_n is the set of actualized vertices and E_n is the set of directed causal edges.

Step 1 — The Open Future. The locus of potential physical interaction is the boundary of the graph: the set of outgoing edges that have not yet terminated at a subsequent vertex. Only events connected to this boundary can be actualized next. The past is fixed; only the frontier is live.

Step 2 — The Quantum Possibility Space. From the boundary ∂G_n, the system explores the space of unactualized candidate interactions. Transition amplitudes are computed via the standard Feynman path integral over the effective background metric $g^{(n)}_{\mu\nu}$ generated by the current graph state. These candidates are virtual: they do not belong to V_n and do not source gravitational curvature. This is why the vacuum does not gravitate in RA — unactualized potentia carry no metric weight.

Step 3 — The Kinematic Snap. Not all candidates can become permanent facts. A candidate vertex v_* is promoted to the valid actualization set $\mathscr{V}_{\text{valid}}$ only if it satisfies the actualization criterion: irreversible increase in entanglement entropy with the asymptotic environment. In the perturbative regime this is implemented by the on-shell threshold $p^\mu p_\mu \geq m^2c^2$. This is the objective, observer-independent collapse mechanism — not an axiom but a consequence of the process ontology.

Step 4 — The Stochastic Update. The universe advances by selecting exactly one vertex from $\mathscr{V}_{\text{valid}}$, weighted by the Born rule measure $W(v_*) = |\mathscr{A}(v_*)|^2$:

$$P(v_{n+1}) = \frac{W(v_{n+1})}{\sum_{v_i \in \mathscr{V}_{\text{valid}}} W(v_i)}. \tag{1}$$

The graph updates: $V_{n+1} = V_n \cup \{v_{n+1}\}$. All contradictory virtual histories branching from the same causal antecedents are pruned. This is wave function collapse — not as a postulate, but as the inevitable consequence of a single irreversible addition to the causal ledger.

Step 5 — The Hydrodynamic Limit (Emergent GR). Gravity is not mediated by a particle during Step 2. It is the large-scale statistical geometry of the actualized graph, recalculated *only after* Step 4 from the updated vertex set. The local actualization density $\lambda(x)$ is computed over a mesoscopic coarse-graining volume, and the effective spacetime metric updates via the RA-modified field equations (Section 8.3). This new metric then serves as the background for Step 2 of the next iteration. Gravity and quantum mechanics operate on different steps of the same algorithm — which is precisely why they have been so resistant to unification by frameworks that try to run them simultaneously.

The BMV prediction. Because the spacetime metric is recalculated strictly in Step 5 — predicated entirely on the set of actualized vertices V_n — it is formally impossible for a quantum superposition (Step 2) to source a superposed gravitational field. Relational Actualism therefore predicts a strict null result for experiments attempting to observe gravity-mediated entanglement between superposed masses (the Bose-Marletto-Vedral protocol [Bose et al., 2017, Marletto and Vedral, 2017]). Spacetime cannot be caught in a superposition because spacetime is the accumulated ledger of interactions that have already irreversibly snapped. This is the most near-term discriminating prediction of the full Engine of Becoming framework.

How RAGC's results fit the algorithm. Every result in this paper is a consequence of one or more steps of this algorithm. The Local Ledger Condition (Section 5) is the conservation law enforced by the Step 4 pruning. The Actualization Projector $\mathscr{P}_{\text{act}}$ (Section 8.3) is the formal implementation of the Step 3 threshold. The metric sourcing equation (Derivation 5, Section 8.5) is the Step 5 hydrodynamic limit. The vacuum energy suppression (Theorem 8.4) follows from the Step 2 non-sourcing of virtual states. The Hubble tension prediction (Prediction 1) arises from the spatial variation in the Step 5 coarse-graining rate. RAGC is not a collection of separate results; it is the first systematic derivation of the consequences of a single algorithm.

3 Postulates and Definitions

Postulate 1 (Ontic State)**.** The fundamental state of a physical system is described by its density operator ρ on a separable Hilbert space $\mathscr{H}$.

Postulate 2 (Actualization)**.** Actualization is the objective, irreversible suppression of off-diagonal coherences in ρ, driven by environmental entanglement. This criterion runs parallel to dynamical collapse models [Bassi et al., 2013], with the key distinction that the trigger in RA is a specific on-shell boson exchange condition rather than a stochastic gravitational threshold. The actualization criterion—irreversible increase in entanglement entropy with the asymptotic environment—is *sufficient* for actualization in all regimes. Its necessity in full generality, and its extension to gauge theories with non-trivial edge mode structure, are open problems inherited from RAQM [Sandeman, 2026a].

Postulate 3 (The Environment)**.** The environment driving actualization is the coarse-grained geometry of the spacetime manifold itself.

Postulate 4 (Bandwidth Limit)**.** The manifold possesses a fundamental information-theoretic channel capacity. There exists a maximum rate $R_{\max}$ at which the local spacetime geometry can process quantum state transitions.

Postulate 5 (Actualization Projector)**.** The macroscopic metric is sourced exclusively by on-shell interactions. Off-shell virtual fluctuations are mathematically unactualized potentia. We formally propose a projector $\mathscr{P}_{\text{act}}$ that restricts the quantum stress-energy tensor to states satisfying $p^\mu p_\mu = m^2c^2$. The precise definition of this projector, and the proof that it suppresses the vacuum energy contribution, is developed in Section 8.3.

Remark 3.1. Unlike Postulates 1–4, which state ontological commitments, Postulate 5 constitutes a conjecture whose formal elaboration is an open mathematical problem. We state it here to motivate the framework; the mathematical development in Section 8.3 constitutes the first rigorous partial derivation.

4 Proper Time and Relativistic Dilation in the Actualization Framework

4.1 Proper Time as Actualization Count

The identification of time with the growing causal graph has a precise corollary for the meaning of proper time. In standard GR, proper time is the spacetime interval along a worldline:

$$d\tau^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu. \tag{2}$$

In RA, the spacetime metric is emergent from the causal graph rather than fundamental. The primitive quantity is not the interval but the count of actualization events along the worldline. We define:

Definition 4.1 (Proper Actualization Count)**.** The *proper actualization count* $\mathscr{N}(\gamma)$ of a worldline γ is the number of on-shell actualization events that occur along γ between two events $p, q \in \mathscr{G}$.

The proper time measured by a physical clock is proportional to $\mathscr{N}(\gamma)$: a clock is a physical system that accumulates a record of its own actualization history. In the continuum limit (established in Derivation 2, Section 8.2), $\mathscr{N}(\gamma)$ converges to the standard GR proper time integral along the emergent metric. The identification is not postulated; it is a consequence of the continuum limit of the RA-CSG dynamics.

4.2 Special-Relativistic Time Dilation

Consider two observers $\mathscr{O}$ and $\mathscr{O}'$ in relative motion, with $\mathscr{O}'$ moving at velocity v relative to $\mathscr{O}$. In the RA account, each observer accumulates actualization events at a rate determined by the on-shell interactions along their respective worldlines.

The actualization criterion is Lorentz-invariant: whether a given virtual exchange constitutes an actualization event is determined by the scalar condition $p^\mu p_\mu = m^2c^2$, which is unchanged by any Lorentz transformation. However, the *rate* at which such events accumulate along a given worldline is frame-dependent, because the Lorentz boost changes the energy-momentum of mediating bosons:

$$p^\mu \longrightarrow \Lambda^\mu{}_\nu\, p^\nu, \tag{3}$$

where Λ is the Lorentz transformation. In the rest frame of $\mathscr{O}'$, the typical boson energy is $\langle E \rangle$. In the frame of $\mathscr{O}$, the same bosons are blue- or red-shifted; the threshold crossing rate — and therefore the actualization rate per coordinate second — changes accordingly.

The ratio of proper actualization counts for the two observers between the same pair of events is:

$$\frac{\mathscr{N}(\gamma')}{\mathscr{N}(\gamma)} = \sqrt{1 - \frac{v^2}{c^2}} = \frac{1}{\gamma_{\text{LF}}}, \tag{4}$$

where $\gamma_{\text{LF}} = (1 - v^2/c^2)^{-1/2}$ is the Lorentz factor. This is the standard time dilation formula, now derived from the actualization count rather than from the metric interval. The moving clock accumulates fewer actualization events per coordinate second not because time "flows more slowly" in some absolute sense, but because its worldline through the causal graph traverses a different density of on-shell threshold crossings.

Remark 4.2. This account makes the origin of SR time dilation transparent in a way that the geometric account does not. The geometric account says: the spacetime interval is shorter along the moving worldline. The RA account says: the moving observer undergoes fewer irreversible causal interactions per coordinate second. The second is more fundamental, because it gives a physical mechanism (actualization rate) for what the first describes geometrically. The two accounts agree quantitatively in the continuum limit; the RA account is more primitive.

4.3 Gravitational Time Dilation

Gravitational time dilation in RA follows from the metric sourcing equation rather than from the Lorentz boost. A region of high actualization density $\lambda(x)$ corresponds to a densely woven causal graph — many vertices per unit coordinate volume, tightly constrained by existing causal relations. A clock embedded in such a region is embedded in a dense causal structure; each new actualization event it undergoes is constrained by many prior events.

In the weak-field limit, the gravitational potential Φ is sourced by $\lambda(x)$ via the RA Poisson equation:

$$\nabla^2\Phi = 4\pi G\rho_A(x), \qquad \rho_A \propto \frac{A_{\mathrm{RA}}(x)}{V}, \tag{5}$$

where ρ_A is the accumulated causal depth density. A clock at potential Φ accumulates actualization events at a rate modified by the local graph topology. In the Newtonian limit:

$$\frac{\mathcal{N}(\gamma_\Phi)}{\mathcal{N}(\gamma_0)} \approx 1 + \frac{\Phi}{c^2}, \tag{6}$$

where γ_0 is a reference worldline far from the source and γ_Φ is a worldline at potential $\Phi < 0$. Clocks deep in a gravitational well accumulate fewer actualization events per coordinate second — they run slower — because the dense causal graph in that region constrains the rate at which new on-shell events can be added to the local subgraph.

Unification. Both kinematic and gravitational time dilation are, in RA, the same phenomenon: the rate of accumulation of irreversible causal events along a worldline. The Lorentz boost changes this rate by transforming boson four-momenta; the gravitational field changes it by modifying the local causal graph density. Special-relativistic and general-relativistic time dilation are two regimes of the same underlying actualization-rate calculation.

Hard wall. The formal derivation of equations (4) and (6) from the covariant RA field equations requires establishing Bianchi compatibility ($\nabla_\mu \langle T^{\mu\nu}_{\mathrm{RA}} \rangle = 0$) on general curved backgrounds — Derivation 3, hard wall iv. The qualitative account above is derivable from the actualization criterion and the proper-time identification; the precise numerical coefficients require the full covariant framework. What is not conjectural is the identification of proper time with the actualization count: this follows directly from the ontological commitment that time is the growing graph.

5 The Local Ledger: Conservation on a Growing Graph

In RA, the universe is an actively growing causal graph. Extending the topology of Causal Set Theory [Dowker, 2005] and its classical sequential growth dynamics [Rideout and Sorkin, 2000], new actualization events continuously add vertices and directed edges. Conservation laws in RAGC must be reconceived as local ledger properties rather than global constants.

Define the actualization graph $G = (V, E)$, where V is the set of actualization events and E is the set of directed causal edges. Each vertex $v \in V$ carries a charge vector $\mathbf{q}(v) \in \mathbb{R}^n$, whose components are the conserved charges (energy-momentum, electric charge, baryon number, etc.).

Definition 5.1 (Local Ledger Condition (LLC))**.** For every vertex $v \in V$:

$$\sum_{\substack{u \in V \\ (u,v) \in E}} \mathbf{q}(u) = \sum_{\substack{w \in V \\ (v,w) \in E}} \mathbf{q}(w). \tag{7}$$

That is, the sum of incoming charges equals the sum of outgoing charges at every vertex. In the continuum limit of a dense graph (vertex density $\lambda \to \infty$), the LLC converges to the standard Noether continuity equation $\nabla_\mu J^\mu = 0$. A sketch of this convergence is given in Section 8.2.

6 Causal Severances and Cosmological Expansion

6.1 Event Horizons and Ledger Partitioning

In RAGC, an event horizon acts as an absolute actualization boundary, topologically severing the causal graph. The Graph Cut Theorem (Theorem 8.9) establishes rigorously that the local conservation condition holds independently on each side of the horizon cut.

Scope clarification. This result resolves the *energy bookkeeping* version of the Hawking radiation problem—specifically, the question of whether total energy is conserved across the horizon. In RAGC, the question "is $Q_{\text{interior}} + Q_{\text{exterior}}$ conserved?" is not well-posed: after the causal cut, Q_{interior} is inaccessible to the exterior subgraph by definition of an event horizon. The claim is *not* that the full unitarity question (whether quantum information is preserved) is resolved—that requires the complete quantum gravity treatment of $\mathscr{P}_{\text{act}}$ in the non-perturbative regime, which is an explicit open problem (Section 8.3.6). This distinction is maintained throughout.

When the actualization flux approaches the bandwidth limit $R_{\max}$, a topological severance occurs, nucleating a causally orthogonal region. This is not merely an analogy with black hole formation: it is a precise claim about what happens when the local rate of on-shell interactions saturates the information-processing capacity of the spacetime manifold. The DAG can no longer weave new edges into the existing causal structure fast enough to maintain connectivity; a new, topologically disconnected subgraph is nucleated.

The ongoing accretion from the parent spacetime feeds an external energy flux into this daughter region through the severance boundary. In RAGC, this flux is not arbitrary — it is governed by the Local Ledger Condition at the boundary vertices (Theorem 8.9), which constrains the charge flow across the cut. The energy density that the daughter spacetime inherits at nucleation depends on the actualization flux at the moment of severance, which is determined by the local matter density and interaction rates in the parent spacetime at that moment.

This provides a candidate mechanism for cosmological constant selection: different daughter spacetimes, nucleated at different moments and locations in the parent DAG, inherit different initial energy densities from the boundary flux. The effective cosmological constant of each daughter spacetime is thus not a fundamental free parameter but a derived quantity — selected by the actualization conditions at the moment of causal severance. Spacetimes nucleated from high-flux regions inherit higher initial energy densities; those from low-flux regions inherit lower ones.

Relationship to the anthropic landscape. This mechanism offers a physically grounded alternative to the string theory landscape as an account of cosmological constant selection. In the landscape picture, the cosmological constant is selected anthropically from an enormous ensemble of vacua. In RAGC, the selection mechanism is dynamical and local: the value of Λ in a daughter spacetime is determined by the actualization flux at its nucleation event, not by a random draw from a distribution of vacua. This is a sharper prediction in the sense that it ties the cosmological constant to measurable properties of the parent spacetime's matter distribution and interaction rates.

We note this entire discussion as a conjecture, not a derivation. The quantitative prediction — establishing the functional relationship between the boundary flux at nucleation and the effective

Λ of the daughter spacetime — requires the non-perturbative treatment of $\mathscr{P}_{\text{act}}$ (Section 8.3.6) and the resolution of the Bianchi compatibility problem. It is, however, a *testable* conjecture in principle: if our universe is a daughter spacetime in this sense, its cosmological constant should reflect the actualization flux conditions at the nucleation event of our causal structure, which in turn should leave an imprint in the large-scale structure of the early universe. Identifying that imprint is a long-range observational target for the program.

6.2 Intrinsic Graph Growth and the Hubble Tension

Independent of extrinsic feeding, the universe experiences intrinsic graph growth. As matter and energy interact, the continuous actualization of on-shell events physically weaves new topological edges. Actualization-dense regions (galaxy clusters) have high edge density; their metric is, in the continuum limit, tightly constrained. Intergalactic voids, with low actualization rate density $\lambda(x)$, permit maximum intrinsic topological expansion.

The mechanism rests on a distinction between the *rate of vertex addition* and the *coordinate volume subtended per vertex*. In dense regions, vertices are added rapidly but each is tightly coupled to neighbors, leaving little room for the metric to expand between events. In voids, vertices are added rarely, but each subtends a large coordinate volume because no neighboring interactions constrain the local geometry. The metric breathes freely between sparse causal stitches in a way it cannot in a densely woven region.

This generates the prediction $H_{\text{eff}}(x) \propto 1/\lambda(x)$, which provides the mechanism for the Hubble Tension. The full statement of this prediction, its distinguishing observational signature, and its distinction from ΛCDM and early dark energy models are given in Section 7, Prediction 1. The quantitative derivation requires fixing the generative measure of the DAG (Section 8.1).

7 Empirical Predictions

The framework makes predictions that are testable observationally and experimentally, independently of whether every mathematical derivation in Section 8 is complete. This section states them plainly. A theory earns its standing by making predictions, not by deferring all claims until every derivation is polished. The quantitative precision of each prediction depends on completing the derivations identified in Section 8; the qualitative direction and distinguishing signature of each prediction follows from the core postulates alone.

7.1 Prediction 1: The Hubble Tension as a Density-Dependent Expansion Rate

RAGC predicts that the effective cosmic expansion rate $H_{\text{eff}}(x)$ is inversely proportional to the local actualization density $\lambda(x)$:

$$H_{\text{eff}}(x) \propto \frac{1}{\lambda(x)}. \tag{8}$$

This is a direct structural consequence of the DAG metric: in regions of high actualization density (galaxy clusters, filaments), the metric is tightly woven and resists expansion; in regions of low actualization density (intergalactic voids), the metric grows freely between sparse causal events.

The Hubble Tension. Early-universe probes (CMB temperature anisotropies, baryon acoustic oscillations) sample the mean actualization rate of the dense, radiation-dominated early universe. Late-universe probes (Cepheid distance ladders, Type Ia supernovae, strong lensing time delays) are dominated by lines of sight through low-λ void-dominated regions. RAGC predicts these two probe families measure genuinely different physical quantities: H_{eff} in different density regimes. The persistent $\sim$4–6σ discrepancy between early and late measurements is therefore not a systematic error but a real signal of density-dependent expansion.

Distinguishing signature. This prediction is observationally discriminating in the following sense: the inferred H_0 from any probe should correlate with the mean matter density along its line of sight. Void-dominated survey geometries should return systematically higher H_0; cluster-dominated geometries should return lower values. This line-of-sight dependence is *not* predicted by ΛCDM (which treats H_0 as a global constant) nor by competing early dark energy models (which shift the sound horizon uniformly). RAGC predicts a specific functional relationship between H_0 and the integrated mass density along each line of sight, quantifiable once the generative measure of the DAG is fixed (Section 8.1).

No dark energy required. The differential expansion rate between dense and sparse regions is a structural consequence of the actualization ontology itself, requiring no additional dark energy ingredient.

7.2 Prediction 1b: The Bandwidth-Partition Formula and the Eridanus Test

The mechanism of Prediction 1 can be made quantitative without requiring the full derivation of the generative measure c_k. The total actualization bandwidth is partitioned between matter-binding edges and spatial-expansion edges:

$$\lambda_{\text{total}} = \lambda_m + \lambda_{\text{exp}}, \tag{9}$$

where λ_m is the rate of causal connections that bind matter (internal knots of the causal graph) and λ_{exp} is the rate of connections that grow the graph spatially (transverse expansion edges). Because the total bandwidth is locally conserved, a deficit in matter density frees bandwidth for expansion. For a region with fractional density contrast $\delta \equiv (\rho - \bar{\rho})/\bar{\rho}$ relative to the cosmic mean, the local expansion bandwidth is:

$$\lambda_{\text{exp}}^{\text{local}} = \bar{\lambda}_{\text{exp}} - \delta\,\bar{\lambda}_m, \tag{10}$$

giving the ratio of local to global Hubble rates:

$$\frac{H_{\text{local}}}{\bar{H}} = 1 - \delta\,\frac{\bar{\lambda}_m}{\bar{\lambda}_{\text{exp}}}. \tag{11}$$

One-parameter fit to the KBC void. The KBC void [Kenworthy et al., 2019] is a well-documented underdensity of radius ∼300 Mpc with density contrast $\delta \approx -0.20$. Using the SH0ES local measurement $H_{\text{local}} = 73.04 \pm 1.04\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ [Riess et al., 2022] and the Planck CMB value $\bar{H} = 67.4\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, the implied ratio is:

$$\frac{\bar{\lambda}_m}{\bar{\lambda}_{\text{exp}}} = \frac{H_{\text{local}}/\bar{H} - 1}{-\delta} = \frac{0.084}{0.20} \approx 0.42. \tag{12}$$

This is a one-parameter fit. The ratio $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}$ will be derived from first principles once the BDG-filter MCMC (Section 8.1, Hard Wall 1) is complete; until then it is observationally calibrated from the KBC void.

Falsifiable prediction: the Eridanus supervoid. The Eridanus supervoid has an estimated density contrast $\delta \approx -0.30$ and diameter ∼300 Mpc [Naidoo et al., 2016]. Substituting into equation (11) with the fitted ratio:

$$H_{\text{Eridanus}} = 67.4 \times \big(1 - (-0.30)(0.42)\big) \approx 75.9\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}. \tag{13}$$

This prediction is RA-native and structurally distinct from ΛCDM (which predicts no dependence of H_0 on the density of the observing environment) and from standard void models (which produce at most ∼1–2 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ corrections from peculiar velocities alone). A measurement of H in the direction of the Eridanus supervoid consistent with ∼75.9 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ would provide strong support for the bandwidth-partition mechanism; a value near 67–68 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ would falsify it.

Linear dependence on δ: the full observational test. Equation (11) predicts that H_0 measurements should vary *linearly* with the mean density contrast $\bar{\delta}$ along each line of sight, with slope $-\bar{H} \cdot (\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}) \approx -28.3\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ per unit δ. This is a falsifiable functional form, not merely a direction. Current data from the Carnegie-Chicago Hubble Program, H0LiCOW, and TDCOSMO, stratified by line-of-sight density, already contain sufficient statistical power to test this slope prediction against the ΛCDM null of zero slope.

7.3 Prediction 2: Vacuum Energy Suppression and CMB Implications

RAGC predicts that the vacuum energy contribution to the metric source term is structurally zero:

$$\langle 0 | T^{\mu\nu}_{\text{RA}} | 0 \rangle = 0, \tag{14}$$

as proved in Theorem 8.4. At the ontological level, this is a structural consequence of the actualization postulate: the metric is sourced only by actualized (on-shell) states, and the vacuum contains none. At the operational level in semiclassical gravity, this motivates the vacuum-subtraction prescription (equation (31) in Section 8.3), which implements the same suppression covariantly within the renormalized framework. The two framings are not in tension: the ontological claim explains *why* the vacuum does not gravitate in RA; the prescription specifies *how* to compute this consistently in the presence of renormalization. Neither is a fine-tuning.

Distinction from ΛCDM. In ΛCDM, a cosmological constant Λ is inserted by hand to account for the observed accelerated expansion, with its value fine-tuned to $\rho_\Lambda \sim 10^{-47}$ GeV^4 against the QFT vacuum prediction of $\sim 10^{74}$ GeV^4. RAGC eliminates the $\sim 10^{120}$ discrepancy structurally: the vacuum does not gravitate because the vacuum contains no actualized states. The effective expansion of the universe is then driven entirely by the intrinsic graph growth mechanism of Prediction 1, not by vacuum energy.

Observational target. If vacuum energy does not contribute to the metric source, the CMB power spectrum should reflect a universe whose expansion history is governed entirely by matter and radiation actualization densities, without a Λ term. The angular power spectrum, particularly at low multipoles where dark energy effects are most pronounced, should deviate from ΛCDM predictions in a manner calculable from the RA-modified Einstein equation $G_{\mu\nu} = 8\pi G \langle \psi_{\text{phys}} | T^{\mu\nu}_{\text{RA}} | \psi_{\text{phys}} \rangle$. Quantifying this deviation is a concrete observational target for the program, pending the Bianchi compatibility derivation (Section 8.3.6).

7.4 Prediction 3: Cosmological Constant Selection in Daughter Spacetimes

When the actualization flux approaches the bandwidth limit $R_{\max}$, a topological severance nucleates a causally disconnected daughter spacetime. The Ledger Partition Theorem (Theorem 8.9) establishes that the Local Ledger Condition holds independently on each side of the cut. The energy density inherited by the daughter spacetime at nucleation is governed by the boundary flux at that moment, which is determined by the local matter density and interaction rates in the parent spacetime.

The prediction. RAGC predicts that the effective cosmological constant Λ_{eff} of a daughter spacetime is not a fundamental free parameter but a derived quantity:

$$\Lambda_{\text{eff}} \sim \Phi_{\text{boundary}}(C^*), \tag{15}$$

where $\Phi_{\text{boundary}}(C^*)$ is the total actualization flux crossing the severance boundary C^* at the moment of nucleation. Spacetimes nucleated from high-flux regions inherit higher Λ_{eff}; those from low-flux regions inherit lower values.

Distinction from the anthropic landscape. In the string theory landscape, Λ is selected from an enormous statistical ensemble of vacua, with the observed value explained anthropically. In RAGC, the selection mechanism is dynamical and local: Λ_{eff} is determined by a specific, in-principle-measurable physical quantity (the boundary flux) at a specific event (the nucleation). This is a sharper, more falsifiable account: it ties the cosmological constant to the large-scale structure and matter distribution of the parent spacetime at the nucleation epoch.

Observational imprint. If our universe is a daughter spacetime in this sense, the nucleation event should leave an imprint in the large-scale structure of the early universe—specifically, in correlations between the matter distribution at the nucleation epoch and the effective cosmological constant inferred from late-time observations. Identifying this imprint is a long-range observational target.

Connection to biological complexity. The causal graph mechanisms of RAGC extend directly to the emergence of biological complexity. The Local Ledger Condition and Ledger Partition Theorem, proved here, ground the conservation structure that biological assembly inherits. The companion paper RACI [Sandeman, 2026c] establishes Condition C1′: each JOIN operation in Assembly Theory's molecular construction program corresponds to at least one irreversible actualization event, operationally defined as an entropy-producing on-shell energy transfer not reversed by subsequent unitary evolution. This coarse-grained lower bound is supported by a variational argument and a B3LYP/6-311+G* computational survey across the five primary bond types of biological assembly (C-C, C-N, C-O, S-S, P-O), all confirmed exothermic. The physics that generates the DAG metric is the same physics that makes biological complexity possible. The further extension to agentic intelligence— formalizing how biological systems develop generative models of their causal environment and use them to steer the growth of the DAG—is developed in the companion paper RAHC [Sandeman, 2026d].

7.5 Prediction 4: The Dark Matter Signature as an Actualization-Density Effect

A particle species that undergoes no on-shell electromagnetic or strong-force interactions contributes negligible actualization flux to the causal DAG and therefore negligible metric sourcing. RAGC predicts that Weakly Interacting Massive Particles (WIMPs) cannot account for galactic gravitational lensing, and that the observed "dark matter" signatures have two distinct actualization-density explanations requiring no new particle species.

The WIMP Prohibition. For a canonical 100 GeV WIMP with spin-independent cross-section $\sigma_\chi \lesssim 10^{-44}$ cm^2 (consistent with current LZ and XENON1T bounds), the actualization rate in the Milky Way ISM ($n_b \sim 1$ nucleon cm^{-3}, $v \sim 200$ km s^{-1}) is:

$$\Gamma_\chi = n_b \cdot \sigma_\chi \cdot v \;\approx\; 2 \times 10^{-37}\ \text{s}^{-1} \text{ per WIMP.} \tag{16}$$

The mean time between actualization events is $\tau = 1/\Gamma_\chi \approx 5 \times 10^{36}$ s, which is $\sim 10^{18}$ times the age of the universe. A WIMP traverses the causal DAG without writing vertices.

By contrast, a baryonic proton in the same environment undergoes electromagnetic actualization at:

$$\Gamma_p = n_b \cdot \sigma_{\text{EM}} \cdot v_p \;\approx\; 10^{-14}\ \text{s}^{-1}, \tag{17}$$

using a conservative atomic Coulomb cross-section $\sigma_{\text{EM}} \sim 10^{-20}$ cm^2. The ratio of volumetric actualization flux is:

$$\frac{\lambda_{\text{baryon}}}{\lambda_{\text{WIMP}}} \;\approx\; 10^{25}\text{–}10^{38}, \tag{18}$$

where the range reflects different galactic environments (diffuse ISM to stellar-density regions). In RA, the metric is sourced by actualization flux $\lambda(x)$, not rest-mass density. A WIMP halo of five

times the baryonic mass contributes $\lesssim 10^{-25}$ of the metric sourcing of the visible baryonic matter. This is a categorical prohibition, not a quantitative suppression: WIMPs cannot account for observed galactic gravitational lensing in the RA framework.

Flat rotation curves, the Bullet Cluster, and the topological boundary mechanism. The same actualization-density field that produces the WIMP prohibition also suggests mechanisms for flat galactic rotation curves (via a $\nabla^2(\ln\lambda)$ gradient correction to the metric sourcing equation) and for the Bullet Cluster gravitational lensing offset (via the distinction between instantaneous actualization flux λ and accumulated causal depth A_{RA}). These mechanisms are conjectural and require covariant derivation before they constitute a dark matter replacement. The covariant form of the $\nabla^2(\ln\lambda)$ weak-field correction is $\Box(\ln\lambda)$, where $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the covariant d'Alembertian; demonstrating how this couples to $g_{\mu\nu}$ as a sourcing term, recovering the weak-field Poisson equation in the appropriate limit, and establishing Tully-Fisher mass scaling, CMB structure formation, and quantitative Bullet Cluster lensing convergence are the open problems of RADM [Sandeman, 2026f]. They are developed as the primary subject of the companion paper RADM [Sandeman, 2026f].

8 Mathematical Derivations

The following four derivations represent the core mathematical open tasks of RAGC. They are *derivations*—mathematical tasks required to connect the framework's postulates to quantitative predictions—not empirical questions. The predictions of Section 7 stand on the basis of the postulates and proved theorems regardless of whether these derivations are complete. For each derivation, we provide a partial result or proof sketch that takes the formalism as far as currently tractable, identifies the hard wall precisely, and states what a collaborator would need to do to complete the result.

8.1 Problem 1: The Generative Algebra of the Causal Graph

Goal. Define the rigorous algorithmic update rules by which the universe builds its DAG topology, incorporating the on-shell constraint.

8.1.1 Foundation: Classical Sequential Growth Dynamics

The Classical Sequential Growth (CSG) dynamics of Rideout and Sorkin [Rideout and Sorkin, 2000] defines a growing sequence of finite causal sets $\mathscr{C}_0 \subset \mathscr{C}_1 \subset \mathscr{C}_2 \subset \cdots$ via Markov transitions. At each step, one new element e is added with transition probability

$$t\big(\mathscr{C}_n \to \mathscr{C}_n + \{e\}\big) \;=\; \prod_k c_k \cdot \binom{n_k}{\text{coeff}}, \tag{19}$$

where c_k are coupling constants over precursor sets and n_k is the count of elements with that precursor structure. The key property is *causal invariance*: the transition probabilities commute for spacelike-separated additions, so the final distribution is independent of the order in which such additions are made—the discrete analogue of diffeomorphism invariance.

8.1.2 The RA Extension: On-Shell Filter

RAGC augments CSG dynamics with a physical constraint. Each potential new element e is assigned a type label $\tau(e) \in \{\text{on-shell, off-shell}\}$, determined by whether the underlying quantum transition has support on the mass-shell Σ_m (see Section 8.3). The RA-extended transition probability is

$$t_{\mathrm{RA}}\big(\mathscr{C}_n \to \mathscr{C}_n + \{e\}\big) \;=\; t_{\mathrm{RS}}\big(\mathscr{C}_n \to \mathscr{C}_n + \{e\}\big) \cdot \delta_{\tau(e),\,\text{on-shell}}. \tag{20}$$

The on-shell filter is local (per-element), so causal invariance is inherited from Rideout-Sorkin: if two spacelike-separated transitions commute in the unfiltered theory, they commute in the filtered theory because the filter acts independently on each. The adjacent possible at step n is

$$\mathrm{AP}(\mathscr{C}_n) \;:=\; \big\{\mathscr{C}_n \cup \{v\} : \tau(v) = \text{on-shell},\ \exists\,\text{precursor antichain } A \subseteq \mathscr{C}_n\big\}. \tag{21}$$

8.1.3 Process Algebra Formulation

Following Sulis [Sulis, 2020], we express the RA-CSG dynamics as a CCS-like process algebra. The causal set process P is defined by

$$\begin{aligned} P_{\text{causal set}} ::= {} & 0 && \text{(no further on-shell transitions)} \\ & \mid \tau_{\text{on-shell}} . P_{\text{causal set}} && \text{(on-shell transition, then continue)} \\ & \mid P + Q && \text{(non-deterministic choice of next element).} \end{aligned}$$

Actualization corresponds to the communication of two processes (emitter and absorber) along a shared on-shell channel. This formalism makes the bilateral nature of actualization explicit: a vertex nucleates only when both participating systems contribute an on-shell event.

8.1.4 Hard Wall: The Generative Measure and Benincasa-Dowker Filter

The central open problem is fixing the coupling constants c_k in the RA-CSG transition probabilities. In standard Rideout-Sorkin CSG, these are genuinely free parameters [Sorkin, 2003]. RA provides a principled discrete constraint that vanilla CST lacks: the actualization filter.

The Benincasa-Dowker sign criterion. The Benincasa-Dowker-Glaser (BDG) action [Benincasa & Dowker, 2010] provides the unique discrete, retarded, Lorentz-invariant action functional for a causal set. At a candidate vertex v_* in 4D, it is pure graph combinatorics:

$$S^{(4)}_{\text{BDG}}(v_*) = \frac{1}{l_P^2}\big(1 - N_1(v_*) + 9N_2(v_*) - 16N_3(v_*) + 8N_4(v_*)\big), \tag{22}$$

where $N_k(v_*)$ counts the k-element inclusive causal intervals in the immediate past of v_*. No metrics or PDEs appear; only the combinatorial structure of the causal past is used. In the continuum limit, the mean of S_{BDG} recovers the Einstein-Hilbert action [Benincasa & Dowker, 2010].

The RA actualization criterion $\Delta S(\rho\|\sigma_0) > 0$ has a natural discrete counterpart via the BDG action: a candidate vertex v_* actualizes if and only if

$$S^{(4)}_{\text{BDG}}(v_*) > 0. \tag{23}$$

This identification is RA-native: a vertex contributes positively to the local scalar curvature if and only if it increases the local relative entropy with respect to the vacuum. The sign criterion replaces the phenomenological on-shell filter with a purely graph-theoretic one.

Mean-field approximation and Poisson-CSG. The RA actualization filter, combined with QFT cluster decomposition (spacelike-separated interactions decohere independently), implies that the number of actual causal connections k formed by a new vertex in an environment with actualization density $\lambda(x)$ follows

$$c_k^{\text{RA}}(x) = \frac{e^{-\mu(x)}\,\mu(x)^k}{k!}, \qquad \mu(x) = \lambda(x)\,V_{\text{coh}}(x)\,p_{\text{th}}, \tag{24}$$

where $V_{\text{coh}}(x)$ is the coherence volume of the local quantum state and $p_{\text{th}} \approx 10^{-2}$ is the Kinematic Coherence Bound of RAQI [Sandeman, 2026e]. This Poisson-CSG formula is the mean-field approximation to the exact BDG filter. It makes c_k *local and environment-dependent* — a structural difference from all prior CSG formulations, where c_k are universal constants.

This locality has two immediate consequences. In dense environments ($\mu \gg 1$): c_k peaks at large k, the graph is highly connected, $d_s = 3$, and standard GR is recovered. In sparse environments ($\mu \ll 1$): c_k is dominated by $k = 0$ and $k = 1$, the graph becomes tree-like, and by the Durhuus-Jonsson-Wheater theorem [Durhuus et al., 2010], $d_s = 2$. This is the mechanism that drives dimensional reduction in galactic halos (RADM [Sandeman, 2026f], Section 3.2) and underlies the conditional Tully-Fisher derivation.

Remaining gap. Two quantities require determination before the Poisson-CSG formula yields quantitative predictions:

1. The coherence volume $V_{\text{coh}}(x)$: the volume of the pre-actualization quantum state (the off-shell entanglement region that collapses upon a Kinematic Snap). This is not the Planck volume but depends on the local energy scale and quantum state structure.

2. The exact c_k distribution beyond the mean-field limit: the full BDG + RA filter calculation requires MCMC simulation of the RA-CSG growing under equation (23), extracting the asymptotic c_k distribution and the ratio $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}$ (which determines the Hubble tension prediction of Section 7).

Collaborator target: causal set theory, CDT computational physics, quantum information theory.

8.2 Problem 2: The DAG Continuum Limit as an Emergent Shadow

Goal. Prove that the continuous equations of General Relativity are the emergent shadows of the discrete graph-rewriting process in the limit of infinite vertex density.

8.2.1 LLC → Continuity Equation

Working in natural units ($c = \hbar = 1$) throughout this section. Let G be a DAG with vertex density λ (vertices per unit 4-volume, units $[\text{length}]^{-4}$) satisfying the LLC. Define the smoothed charge current

$$J^{\mu}_{\text{smooth}}(x) \;:=\; \frac{1}{\varepsilon^4\,\lambda} \sum_{v:\,|v-x|<\varepsilon} q(v)\,u^{\mu}(v), \tag{25}$$

where $u^{\mu}(v)$ is the unit future-directed tangent to the local causal direction at v. The denominator $\varepsilon^4\lambda$ counts the expected number of vertices in the averaging 4-volume—it is dimensionless—so J^{μ}_{smooth} has units of charge density current in natural units ($[\text{charge}]\cdot[\text{length}]^{-3}$), consistent with the standard 4-current. As $\lambda \to \infty$ and $\varepsilon \to 0$ with $\varepsilon^4\lambda \to$ const, a standard Taylor expansion on the discrete divergence shows

$$\partial_{\mu} J^{\mu}_{\text{smooth}} \;\to\; 0 + \mathscr{O}(1/\lambda), \tag{26}$$

recovering the continuity equation $\nabla_{\mu} J^{\mu} = 0$. The error term $\mathscr{O}(1/\lambda)$ vanishes in the continuum limit. This argument is structurally identical to the Bombelli-Henson-Sorkin recovery of the d'Alembertian from a causal set [Bombelli et al., 2009], adapted to the current vector-field context.

8.2.2 Metric Reconstruction

The Hawking-Malament theorem establishes that on a Lorentzian manifold, the causal structure plus a volume measure uniquely determines the metric (up to conformal factor). In the DAG model: the causal structure is the graph G itself, and the volume measure is the vertex count. Therefore, in principle, the metric is determined by the DAG once the continuum limit is taken.

Whether the RA-CSG dynamics generates a DAG that, in the continuum limit, satisfies $G_{\mu\nu} = 8\pi G\,\langle T^{\mu\nu}_{\text{RA}}\rangle$ rather than some other metric is the core open question. Causal Dynamical Triangulation (CDT) [Ambjørn et al., 2012] provides a benchmark: CDT recovers 4D de Sitter spacetime from a purely combinatorial path integral, establishing that the continuum recovery program is achievable from discrete causal structures. The analogous result for RA-CSG is *conjectured but not proven*.

8.2.3 The Unruh Reconciliation: Precise Problem Statement and Resolution Strategy

The Unruh effect presents a conceptual tension for RA. An accelerated (Rindler) observer detects a thermal bath of particles in the Minkowski vacuum. In RA terms: the inertial observer has $\mathscr{P}_{\text{act}}|0\rangle_{\text{Mink}} = 0$ (no on-shell excitations). The Rindler observer sees a thermal state with $\langle T_{\text{act}}\rangle \neq 0$ relative to their vacuum. Does this make actualization observer-dependent?

Why particle-number cannot be the fundamental criterion. The red-team formulation of this objection is precise: if $\mathscr{P}_{\text{act}}$ is defined in terms of Fock-space particle content, it is frame-dependent, because Rindler and Minkowski observers disagree on particle number by definition. This is correct, and it means that the Fock-space definition of $\mathscr{P}_{\text{act}}$ in Section 8.3 must be understood as the weakly-coupled, perturbative limiting expression of a more fundamental criterion—not the criterion itself.

The AQFT resolution: relative entropy as the fundamental criterion. The fundamental actualization criterion in RA is not "a particle is emitted" but rather "an irreversible increase in entanglement entropy between a localised system and its asymptotic environment occurs." In the language of algebraic QFT, this is a statement about *relative entropy with respect to the vacuum state*—a quantity that is diffeomorphism-invariant and observer-independent.

Formally: an actualization event occurs when an interaction induced by the local algebra $\mathscr{A}(\mathscr{O})$ produces a strictly irreversible increase in the relative entropy $S(\rho\|\sigma_0)$, where ρ is the post-interaction state and σ_0 is the vacuum state. Relative entropy is a scalar under diffeomorphisms. A strictly irreversible thermodynamic exchange is an objective, coordinate-independent fact.

The Bisognano-Wichmann theorem [Bisognano and Wichmann, 1975] in algebraic QFT establishes that the Rindler and Minkowski algebras are unitarily equivalent—describing different perspectives on the same physics, not different physics. The Minkowski and Rindler observers may disagree on the *particle content* of an interaction, but they must agree on whether a thermodynamically irreversible event occurred, because irreversibility is encoded in the monotone decrease of relative entropy—a frame-independent quantity. That irreversible event registers as a permanent vertex on the DAG regardless of which observer describes it.

Finite-dimensional proof and remaining AQFT target. The core claim of frame-independence has been proved in finite dimensions and formally verified in Lean 4. Specifically, the theorem `frame_independence` in `RA_AQFT_Proofs_v10.lean` establishes that $S(U\rho U^\dagger\|\sigma_0) = S(\rho\|\sigma_0)$ for any unitary U representing a Lorentz boost, where the proof uses unitary invariance of relative entropy together with the Poincaré-invariance of the vacuum as a physics axiom. The key lemma `log_unitary_conj`—that $\log(UMU^\dagger) = U\log(M)U^\dagger$ for Hermitian M—is a one-line corollary of `Matrix.cfc_conj_unitary`, the general CFC unitary-conjugation result from the Lean-QuantumInfo library [Meiburg et al., 2025].

The remaining target at the full AQFT level is the infinite-dimensional extension: mapping the finite-dimensional relative entropy proof to Araki's modular relative entropy [Araki, 1976] for type III_1 von Neumann algebras, where standard density matrices and the trace definition of relative entropy do not exist. The precise target is to prove that the Rindler KMS state exhibits $\Delta S_{\text{rel}} = 0$ in the sense of Araki's relative entropy on the wedge algebra $\mathscr{A}(W)$, and that $\mathscr{P}_{\text{act}}$ defined on the local net commutes with the Tomita-Takesaki modular flow σ_s^Ω of the vacuum state Ω. This is the exact gap between the finite-dimensional Lean result and a rigorous QFT theorem. **A collaborator with expertise in Tomita-Takesaki modular theory and Araki's relative entropy would be well-positioned to complete this extension.**

The stationarity criterion: why the Rindler thermal state does not actualize. The Kinematic Snap provides a precise answer to the Unruh question that does not require the full AQFT proof. A uniformly accelerated observer sees the Minkowski vacuum as a thermal bath at temperature $T_U = \hbar a/(2\pi c k_B)$. In the correct algebraic QFT language, the relevant object is the *restriction of the Minkowski vacuum to the Rindler wedge algebra*—a KMS state with respect to boost automorphisms (the Bisognano-Wichmann theorem [Bisognano and Wichmann, 1975]). This KMS (thermal) state is in equilibrium with respect to modular flow: its relative entropy with respect to the vacuum reference state is *constant in time* ($\Delta S_{\text{rel}} = 0$). No irreversible increase in relative entropy is occurring. No net

on-shell bosons are being emitted into the asymptotic environment at a rate exceeding absorption. The Rindler KMS state is the universe's stationary potentia as seen by the accelerated observer—a persistent configuration of the quantum field, not a sequence of actualization events.

Both inertial and accelerated observers therefore agree on the causal fact: in the Minkowski vacuum, no actualization is occurring. They disagree only on the modal decomposition (which modes to call "particles")—a difference of description, not a difference of physical reality. Actualization requires an irreversible *increase* in relative entropy, not merely non-zero relative entropy.

The non-unitary element. A natural question arises: since unitary evolution preserves relative entropy, how can $\Delta S_{\text{rel}} > 0$ ever occur? The answer is that actualization is precisely the non-unitary element. The Kinematic Snap is an open-system event: the on-shell boson propagates to the asymptotic environment, which is traced over in forming the reduced state of the system. This environmental tracing—the coarse-graining implicit in the Local Ledger Condition—is the non-unitary step that permits irreversible entropy increase. In the language of open quantum systems, the actualization map is a CPTP (completely positive trace-preserving) channel, not a unitary: it is precisely the $\mathscr{E}$ in the data processing inequality $S(\mathscr{E}(\rho)\|\mathscr{E}(\sigma)) \leq S(\rho\|\sigma)$. An actualization event is the physical realisation of such a channel, with the environment providing the non-unitary element. Formalising $\mathscr{P}_{\text{act}}$ as a specific CPTP map on the local net $\mathscr{A}(\mathscr{O})$ and proving its modular covariance is the open target identified in D2.

Note on virtual exchanges and relative entropy. A claim appearing in the supporting computational scripts—that "off-shell (virtual) exchanges have $S(\rho_{\text{off-shell}}\|\sigma_0) = 0$"— requires clarification. Virtual exchanges are internal-line bookkeeping in perturbation theory; they are not standalone density operators on the wedge algebra in the QFT sense. The correct statement is operational: interactions that remain kinematically constrained to the virtual (off-shell) regime produce no asymptotic on-shell bosons, no environmental record, and therefore no irreversible increase in $S(\rho\|\sigma_0)$. The absence of a DAG vertex is not equivalent to the system being in a zero-relative-entropy state; it means no irreversible record-forming event has occurred. The relative entropy criterion is meaningful for post-interaction states where a comparison to the vacuum can be made; the "virtual regime" is precisely the regime in which no such comparison is yet applicable.

The finite-dimensional Lean proof of `rindler_stationarity` confirms this picture formally: the theorem establishes that $S(U\rho_R U^\dagger\|\sigma_0) = S(\rho_R\|\sigma_0)$ for any Lorentz boost U applied to the Rindler thermal state, which is precisely the statement that $\Delta S_{\text{rel}} = 0$. The remaining step at the full AQFT level is to extend this result to the infinite-dimensional type III_1 algebra setting, replacing the finite-dimensional relative entropy with Araki's modular relative entropy and proving that the Rindler KMS state is a fixed point of the modular flow in that setting.

8.3 Problem 3: $\mathscr{P}_{\text{act}}$ as a Syntactic Type-Constraint

Goal. Define the Actualization Projector rigorously, prove that it suppresses the vacuum energy contribution to the metric source term, and establish its gauge covariance.

8.3.1 The Standard Vacuum Energy Problem

In standard QFT on a curved background, the Einstein equation takes the form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} \;=\; 8\pi G\,\langle 0|T_{\mu\nu}|0\rangle. \tag{27}$$

The QFT vacuum contribution $\langle 0|T_{\mu\nu}|0\rangle \approx M_{\text{Pl}}^4 \sim 10^{74}$ GeV^4, while observations require $\rho_\Lambda \sim 10^{-47}$ GeV^4—a discrepancy of $\sim 10^{120}$. The standard approach renormalizes by subtracting the vacuum term as a counterterm; this is technically consistent but requires extraordinary fine-tuning [Weinberg, 1989].

8.3.2 *Definition of $\mathscr{P}_{\text{act}}$*

Let $\mathscr{H}$ be the Hilbert space of the quantum field. Define the on-shell subspace

$$\mathscr{H}_{\text{on}} := \left\{ |\psi\rangle \in \mathscr{H} : p^\mu p_\mu |\psi\rangle = m^2c^2|\psi\rangle \right\}. \tag{28}$$

In momentum space, $\mathscr{H}_{\text{on}}$ is supported on the positive-frequency mass hyperboloid $\Sigma_m := \{k^\mu \in \mathbb{R}^{3,1} : k^\mu k_\mu = m^2c^2,\ k^0 > 0\}$.

Definition 8.1 (Actualization Projector)**.** $\mathscr{P}_{\text{act}}$ is the spectral projection onto $\mathscr{H}_{\text{on}}$ (natural units $c = \hbar = 1$ throughout):

$$\mathscr{P}_{\text{act}} := \int_{\Sigma_m} \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |k\rangle\langle k|, \tag{29}$$

where $\omega_{\mathbf{k}} = \sqrt{|\mathbf{k}|^2 + m^2}$ is the on-shell energy (in natural units the dispersion relation is $p^\mu p_\mu = m^2$, so $\omega_{\mathbf{k}}^2 = |\mathbf{k}|^2 + m^2$), and the integration is over the Lorentz-invariant measure on Σ_m. The on-shell subspace definition uses $p^\mu p_\mu = m^2$ throughout; in SI units this reads $p^\mu p_\mu = m^2c^2$ and $\omega_{\mathbf{k}} = \sqrt{|\mathbf{k}|^2c^2 + m^2c^4}/\hbar$. For a multi-species theory: $\mathscr{P}_{\text{act}} = \sum_i \int_{\Sigma_{m_i}} \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k},i}} |k,i\rangle\langle k,i|$.

Proposition 8.2. *$\mathscr{P}_{\text{act}}$ is an orthogonal projection: $\mathscr{P}_{\text{act}}^2 = \mathscr{P}_{\text{act}}$ and $\mathscr{P}_{\text{act}}^\dagger = \mathscr{P}_{\text{act}}$.*

Proof. Each $|k\rangle$ is an eigenstate of the mass operator $p^\mu p_\mu$ with eigenvalue m^2c^2. The projection onto an eigenspace of a self-adjoint operator is idempotent and self-adjoint by the spectral theorem. □

8.3.3 *Vacuum Energy Suppression*

Definition 8.3. The RA-modified stress-energy operator is $T_{\text{RA}}^{\mu\nu} := \mathscr{P}_{\text{act}} T^{\mu\nu} \mathscr{P}_{\text{act}}$ (the compression of $T^{\mu\nu}$ to the on-shell subspace: $\langle\psi|T_{\text{RA}}^{\mu\nu}|\phi\rangle = 0$ unless both $|\psi\rangle$ and $|\phi\rangle$ have support on Σ_m). The modified Einstein equation is

$$G_{\mu\nu} \;=\; 8\pi G \,\langle \psi_{\text{phys}}|\, T_{\text{RA}}^{\mu\nu} \,|\psi_{\text{phys}}\rangle, \tag{30}$$

with no bare Λ term.

Theorem 8.4 (Vacuum Energy Suppression)**.** $\langle 0|\, T_{\text{RA}}^{\mu\nu} \,|0\rangle = 0$.

Proof. The QFT vacuum $|0\rangle$ is the Fock state with no on-shell particles. In the Fock representation, $\mathscr{P}_{\text{act}}|0\rangle = 0$: $|0\rangle$ has no support on Σ_m because it contains no on-shell quanta. Therefore

$$\langle 0|\, T_{\text{RA}}^{\mu\nu} \,|0\rangle \;=\; \langle 0|\, \mathscr{P}_{\text{act}} T^{\mu\nu} \mathscr{P}_{\text{act}} \,|0\rangle \;=\; \langle 0|\, \mathscr{P}_{\text{act}} T^{\mu\nu} \cdot 0 \;=\; 0. \qquad \square$$

At the ontological level, this result is a structural consequence of the RA postulate: the vacuum is a superposition of unactualized potentia, and $\mathscr{P}_{\text{act}}$ removes it from the metric source by construction. It is not a fine-tuning. At the operational level in semiclassical gravity, the same result is implemented by the vacuum-subtraction prescription (Section 8.3, hard wall iv):

$$\langle T_{\text{RA}}^{\mu\nu}\rangle[g,\Phi] := \langle T_{\text{ren}}^{\mu\nu}\rangle[g,\Phi] - \langle T_{\text{ren}}^{\mu\nu}\rangle[g,|0\rangle].$$

These are not competing claims: the ontological framing explains *why* the vacuum does not gravitate in RA; the prescription is the *operational implementation* in the renormalized QFT setting, ensuring that configuration differences (Casimir stresses, binding energies) are correctly preserved. The prescription is locally covariant in the flat-space regime; its extension to curved backgrounds via Hadamard renormalization is an open task (D3 hard wall iv). This is distinct from, and complementary to, the Kaloper-Padilla sequestering mechanism [Kaloper and Padilla, 2014]; a future derivation should examine whether they agree at leading order.

8.3.4 Gauge Covariance

Proposition 8.5. *The on-shell condition $k_\mu k^\mu = g_{\mu\nu} k^\mu k^\nu = m^2 c^2$ is a scalar under diffeomorphisms. Therefore $\mathscr{P}_{\text{act}}$ is well-defined in curved spacetime at each fiber of the cotangent bundle.*

Proof. Under a diffeomorphism $\varphi : M \to M$, both k^μ and $g_{\mu\nu}$ transform as tensors, so their contraction is preserved. The subspace $\mathscr{H}_{\text{on}}$ is therefore defined by a coordinate-invariant condition. □

8.3.5 Connection to the DAG Type-Constraint

Definition 8.6 (Vertex Nucleation Condition)**.** A quantum transition $T : v_{\text{in}} \to v_{\text{out}}$ nucleates a new vertex $v \in V(G)$ if and only if $\exists k^\mu$ such that $k^\mu k_\mu = m^2 c^2$—equivalently, if $\mathscr{P}_{\text{act}} |T\rangle \neq 0$.

Claim 8.7 (Virtual Transitions Do Not Nucleate). Off-shell internal lines in Feynman diagrams satisfy $\mathscr{P}_{\text{act}} |T_{\text{virt}}\rangle = 0$ and therefore contribute no vertices to the causal DAG.

Supporting argument. Internal Feynman lines have propagator $\Delta_F(k) = i/(k^2 - m^2 + i\varepsilon)$ with $k^2 \neq m^2$ generically. They represent superpositions over the full off-shell spectrum, with no support on Σ_m. Only asymptotic (external) states and on-shell resonances (at the propagator pole $k^2 = m^2$) have support in $\mathscr{H}_{\text{on}}$. This is precisely the content of the LSZ reduction formula: physical S-matrix amplitudes are the residues at on-shell poles. The causal DAG is therefore built from S-matrix poles, not from the full off-shell generating functional. □

8.3.6 Hard Wall

Three open sub-problems were identified in prior development of this framework. A fourth, raised by external critique, is added here and is arguably the most fundamental.

(i) **Non-perturbative regime.** The definition in Section 8.3 uses Fock-space structure, which requires a perturbative vacuum. In strongly coupled regimes (QCD confinement, condensed matter critical points), the quasi-particle picture breaks down. A non-perturbative definition—perhaps via algebraic QFT (Haag-Kastler axioms) or the OPE—is required.

(ii) **Renormalization compatibility.** $T^{\mu\nu}_{\text{RA}}$ may require its own renormalization scheme. The standard renormalization of $T^{\mu\nu}$ subtracts the vacuum contribution as a counterterm; $\mathscr{P}_{\text{act}}$ achieves this structurally. The two procedures must be shown to agree at the level of renormalized physical amplitudes. Moreover, even granting that $\langle 0|T^{\mu\nu}_{\text{RA}}|0\rangle = 0$ eliminates the absolute vacuum energy density from the metric source, a complete solution to the cosmological constant problem requires showing how *changes* in vacuum energy—Casimir stresses, binding energy contributions, loop corrections to the effective potential—are treated by the RA-modified source. These are finite, physical, and measurable contributions that gravitate in standard semiclassical gravity; demonstrating that $\mathscr{P}_{\text{act}}$ handles them consistently, or identifying the prediction that deviates from standard GR, is an explicit open task.

The trace anomaly. A related and sharper question concerns the conformal (trace) anomaly: the non-vanishing of $\langle T^\mu_\mu \rangle$ for classically conformally-invariant fields on curved backgrounds, which is essential for deriving Hawking radiation. The RA prescription preserves the trace anomaly because $\mathscr{P}_{\text{act}}$ acts on the *already-renormalized* stress tensor $\langle T^{\mu\nu}_{\text{ren}} \rangle$: the trace anomaly arises from the UV renormalization procedure (point-splitting or dimensional regularization) before the on-shell projection is applied. $\mathscr{P}_{\text{act}}$ then projects onto the on-shell sector of the renormalized theory, inheriting the trace anomaly from that theory. Consequently, the RA prescription does *not* eliminate virtual loop contributions that generate the trace anomaly—it eliminates only the configuration-independent absolute vacuum energy. This is consistent with Hawking radiation: the thermal emission from a black hole is sourced by the trace anomaly (and related vacuum polarization effects) of the renormalized theory, which $\mathscr{P}_{\text{act}}$ preserves.

The cosmological constant suppression and Hawking radiation are therefore not in tension within the RA framework.

(iii) **Back-reaction self-consistency.** The modified metric $g_{\mu\nu}$ (sourced by $T^{\mu\nu}_{\mathrm{RA}}$) feeds back into the definition of $\mathscr{H}_{\mathrm{on}}$ via $g_{\mu\nu}k^\mu k^\nu = m^2c^2$. Establishing self-consistency requires a fixed-point argument.

(iv) **Bianchi compatibility and local conservation.** This was the most fundamental open problem. It has been substantially resolved by identifying the correct RA prescription for the stress-energy source, as detailed below.

The vacuum-subtraction prescription and Casimir benchmark. The resolution rests on a precise distinction. The RA actualization projector $\mathscr{P}_{\mathrm{act}}$ suppresses the *absolute* vacuum energy—the configuration-independent contribution $\sim M_{\mathrm{Pl}}^4$ that is present in all states equally. It does *not* suppress energy *differences* arising from physically actualized boundary conditions, because those differences reflect real causal facts about the system. The Casimir effect is precisely such a difference: $E_{\mathrm{Casimir}} = E(\text{plates}) - E(\text{no plates})$. The plates are actualized objects that genuinely modify the causal graph; their boundary conditions are physically real.

This gives the RA prescription a precise mathematical form:

$$\langle T^{\mu\nu}_{\mathrm{RA}}\rangle[g,\Phi] \;:=\; \langle T^{\mu\nu}_{\mathrm{ren}}\rangle[g,\Phi] \;-\; \langle T^{\mu\nu}_{\mathrm{ren}}\rangle[g,|0\rangle], \tag{31}$$

where g is the spacetime metric, Φ are the matter fields, and $|0\rangle$ is the Minkowski (or de Sitter) vacuum state.

Theorem 8.8 (Bianchi Compatibility — Casimir Geometry)**.** *Under the vacuum-subtraction prescription (31):*

(a) $\langle 0|T^{\mu\nu}_{\mathrm{RA}}|0\rangle = 0$ *by construction.* [Vacuum suppression]

(b) $\nabla_\mu\langle T^{\mu\nu}_{\mathrm{RA}}\rangle = \nabla_\mu\langle T^{\mu\nu}_{\mathrm{ren}}\rangle - \nabla_\mu\langle T^{\mu\nu}_{\mathrm{ren}}\rangle\big|_{|0\rangle} = 0$*, provided the standard renormalized tensor satisfies conservation under the Wald axioms (point-splitting scheme).* [Bianchi compatibility]

(c) *The Casimir stress-energy is preserved exactly:* $\langle T^{\mu\nu}_{\mathrm{RA}}\rangle(\textit{plates}) = \langle T^{\mu\nu}_{\mathrm{Casimir}}\rangle$. [Physical vacuum effects reproduced]

(d) *The modified Einstein equation* $G_{\mu\nu} = 8\pi G\langle T^{\mu\nu}_{\mathrm{RA}}\rangle$ *is consistent:* $\nabla_\mu G^{\mu\nu} \equiv 0$ *and* $\nabla_\mu\langle T^{\mu\nu}_{\mathrm{RA}}\rangle = 0$ *both hold.* [Field equation consistency]

Proof. (a) Self-subtraction. (b) Linearity of ∇_μ and conservation of each term separately under Wald axioms. (c) By definition: $\langle T^{\mu\nu}_{\mathrm{ren}}\rangle(\text{plates}) - \langle T^{\mu\nu}_{\mathrm{ren}}\rangle(\text{Minkowski}) = \langle T^{\mu\nu}_{\mathrm{Casimir}}\rangle$. (d) Follows from (b) and the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$. □

Scope of the theorem. Theorem 8.8 establishes Bianchi compatibility within the flat-spacetime regime. Step (b) relies on the linearity of ∇_μ and the assumption that each term separately satisfies $\nabla_\mu\langle T^{\mu\nu}_{\mathrm{ren}}\rangle = 0$ under the Wald axioms (point-splitting scheme). This is established for free quantum fields on globally hyperbolic spacetimes [Wald, 1994]. On general curved backgrounds, the prescription requires a curved-spacetime renormalization scheme (Hadamard point-splitting rather than heuristic vacuum subtraction), and the numerical benchmark uses ordinary derivatives ∂_μ in flat space rather than the covariant divergence ∇_μ in a non-trivial metric. Two things follow from this:

(i) The flat-space Casimir benchmark is valid and numerically confirms the prescription reproduces all measurable vacuum physics correctly.
(ii) Demonstrating $\nabla_\mu \langle T^{\mu\nu}_{\mathrm{RA}} \rangle = 0$ on a curved background—for example, the Casimir apparatus in a weak gravitational field—requires a full curved-spacetime Hadamard renormalization and constitutes a decisive test case (see D3 checklist in Section 8.3.6).

A numerical benchmark at $d = 1\,\mu$m gives Casimir pressure $\langle T^{zz}_{\mathrm{RA}} \rangle = -1.30 \times 10^{-3}$ Pa, consistent with the experimental value $\approx -1.3 \times 10^{-3}$ Pa. The prescription reproduces all measurable vacuum effects while eliminating the $\sim 10^{120}$ vacuum energy catastrophe.

The emergent metric defence. A deeper resolution follows from the ontological hierarchy of the RA framework. The Einstein field equations are not fundamental in RA; they are the thermodynamic equation of state for the underlying causal graph, in the same sense as Jacobson's thermodynamic derivation of GR [Jacobson, 1995]. The Local Ledger Condition enforces energy-momentum conservation exactly at every discrete actualization vertex. The continuous Bianchi identity $\nabla_\mu T^{\mu\nu} = 0$ is the coarse-grained continuum limit of the LLC, not an independent constraint that the discrete theory must satisfy separately. In this sense, the Bianchi compatibility problem dissolves at the level of the fundamental theory: conservation is built into the graph architecture, and the continuous field equations inherit it in the continuum limit.

Remaining open scope. The vacuum-subtraction prescription assumes the standard $\langle T^{\mu\nu}_{\mathrm{ren}} \rangle[g]$ satisfies the Wald axioms, which hold for free fields and established interacting theories. The Wald ambiguity (freedom to add local conserved geometric counterterms) is fixed uniquely in the flat-space limit by the RA condition $\langle 0 | T^{\mu\nu}_{\mathrm{RA}} | 0 \rangle = 0$. Extending to the full non-perturbative regime requires the algebraic QFT formulation of Derivation 2 (D2). These remaining gaps are the same gaps that exist in standard semiclassical gravity; RA does not introduce new difficulties beyond those already present in QFT on curved spacetime.

8.4 Problem 4: The Graph Cut Theorem and Ledger Partitioning

Goal. Prove that the LLC holds independently on each side of a causal severance, and characterize the implications for event horizons.

8.4.1 Formal Setup

Let $G = (V, E)$ be a DAG satisfying the LLC at every vertex. Let $C \subset E$ be a *cut*—a set of edges whose removal disconnects G into two acyclic subgraphs $G_L = (V_L, E_L)$ and $G_R = (V_R, E_R)$ with $V_L \cap V_R = \emptyset$, $V_L \cup V_R = V$.

By the acyclicity of G (global causal order), edges in C run from V_L to V_R (past to future). Boundary vertices in V_L have outgoing cut-edges only; boundary vertices in V_R have incoming cut-edges only.

8.4.2 Theorem and Proof

Theorem 8.9 (Ledger Partition)**.** *Under the above setup:*

(a) *Each subgraph G_L and G_R satisfies the LLC at all interior vertices (those with no edges in C).*

(b) *At boundary vertices, the LLC holds with cut-edges accounted for as boundary flux.*

(c) *The total charge in each sector is independently conserved.*

Proof of (a). For any interior vertex $v \in V_L$, all incident edges lie within E_L. The LLC at v in G holds by hypothesis, and since all relevant edges are in E_L, this is identical to the LLC for G_L. The argument is symmetric for G_R. □

Proof of (b). For a boundary vertex $v \in V_L$ with outgoing edges into C, the LLC at v in G states

$$\sum_{\substack{u \in V_L \\ (u,v) \in E_L}} \mathbf{q}(u) = \sum_{\substack{w \in V_L \\ (v,w) \in E_L}} \mathbf{q}(w) + \sum_{(v,w) \in C} \mathbf{q}_{\text{cut}}(w). \tag{32}$$

This is the LLC at v partitioned by destination—interior (in E_L) plus boundary flux (in C). □

Proof of (c). Define the charge flow $f : E \to \mathbb{R}^n$ with the LLC as the discrete divergence-free condition. By the discrete analogue of Gauss's theorem on G:

$$\sum_{v \in V_L} \operatorname{div}(f)(v) = \sum_{e \in C} f(e) = \Phi(C). \tag{33}$$

Since $\operatorname{div}(f)(v) = 0$ at all interior vertices (LLC holds), the left-hand sum reduces to contributions from boundary vertices only. The total internal charge is conserved: what enters V_L at sources minus what leaves at sinks equals zero by internal LLC, and the difference is exactly $\Phi(C)$. Therefore Q_L is conserved independently of Q_R. □

8.4.3 Application: Event Horizons

An event horizon in RAGC is a maximal causal cut C^*—one where every future-directed causal path from the interior eventually intersects C^*. By acyclicity of the DAG, this is well-defined.

Corollary 8.10 (Horizon Partition)**.** *Let C^* be an event horizon cut. Then G_{int} and G_{ext} each satisfy the LLC independently. The charge in each sector is separately conserved. The quantity $Q_{\text{int}} + Q_{\text{ext}}$ is not observable from the exterior subgraph, because no edges cross from interior to exterior by definition of a horizon. The question of a globally conserved total energy is therefore not well-posed.*

This result establishes the energy bookkeeping dissolution described in Section 6. As noted there, it does not resolve the unitarity question; that requires the non-perturbative treatment of Section 8.3.6. A further scope note: the LLC result is a statement about charge conservation in the discrete DAG graph. Its translation into the language of full GR/QFT—where "local conservation" is $\nabla^\mu T_{\mu\nu} = 0$ and depends on geometry, boundary terms, and the treatment of traced-out degrees of freedom—requires the Bianchi compatibility work identified in Section 8.3.6(iv). The DAG Corollary is a necessary structural result; showing it implies the correct GR boundary condition is an open task.

8.5 Derivation 5: The Unified Metric Sourcing Equation

Status: relocated to RADM. The conjectural two-component metric sourcing equation $\nabla^2\Phi = 4\pi G[\rho_A + \rho_\lambda]$, the derivation of the $\nabla^2(\ln\lambda)$ gradient term, and the associated rotation curve and Bullet Cluster arguments are developed as the primary subject of the companion paper RADM [Sandeman, 2026f]. That paper provides the detailed derivation candidates, hard walls, and collaborator targets for the dark matter replacement programme. The WIMP prohibition (Prediction 4 above) stands independently of the conjectural mechanisms in RADM.

8.6 Derivation 6: Equivalence Principle Recovery and Operationalization of λ

Goal. Two foundational issues require explicit treatment before the dark matter predictions (Derivation 5) can be considered publication-ready: (i) the recovery of the weak equivalence principle (WEP) for ordinary matter from the actualization-dependent sourcing equation, and (ii) the operational definition of $\lambda(x)$ in terms of observables that would allow the Hubble tension prediction (Prediction 1) to be tested quantitatively.

The equivalence principle tension. The WIMP prohibition rests on the claim that species with negligible electromagnetic and strong cross-sections contribute negligible actualization flux and therefore negligible metric sourcing. A referee correctly identifies that this appears to imply gravitational coupling is not universal: it depends on non-gravitational interaction channels, which conflicts with precision tests of the weak equivalence principle (WEP) unless the framework provides an explicit mechanism for universality recovery.

The resolution rests on the ρ_A term of the unified sourcing the unified sourcing equation $\nabla^2\Phi = 4\pi G[\rho_A + \rho_\lambda]$ (developed in RADM [Sandeman, 2026f]). For any species in thermodynamic equilibrium over cosmological time, the accumulated causal depth satisfies

$$\frac{A_{\mathrm{RA}}(x)}{V} \approx \lambda(x) \cdot t_{\mathrm{cosm}}, \tag{34}$$

where t_{cosm} is the cosmological age of the system. In this limit, $\rho_A \propto \lambda \cdot t_{\mathrm{cosm}} \cdot \langle \Delta E \rangle / c^2 \propto \rho_{\mathrm{matter}}$: the accumulated depth is proportional to ordinary mass-energy, and $\rho_\lambda = 0$ (no gradient). Standard GR is recovered, and the WEP is respected for all matter in equilibrium—neutrons, protons, lead atoms, dark baryons if any. Precision tests of the WEP (Eötvös experiments, lunar laser ranging, atom interferometry) probe ordinary matter in quasi-equilibrium conditions; these are the regime where $\rho_A \propto \rho_{\mathrm{matter}}$ exactly. Predicted outcome of torsion-balance tests: standard GR to current precision.

WIMPs violate the universality only in the following precise sense: they have $\lambda_{\mathrm{WIMP}} \approx 0$ at all times (not merely in the current epoch), so $A_{\mathrm{RA,WIMP}} \approx 0$ even over cosmological time. They have never been in actualization equilibrium with the baryonic sector. They are not a species that violates universality; they are a species that has never participated in the actualization dynamics that generate the metric in the first place.

Hard wall: formal proof of WEP recovery. The argument above establishes the equilibrium limit qualitatively but does not constitute a proof. A publishable version of the WIMP prohibition requires:

1. A formal derivation showing that $\rho_A \to \rho_{\mathrm{matter}}$ in the equilibrium limit, with explicit bounds on the correction term $|\rho_A - \rho_{\mathrm{matter}}|/\rho_{\mathrm{matter}}$ as a function of departure from equilibrium.
2. An explicit prediction for fifth-force experiments and atom interferometry: any species with $\lambda \ll \lambda_{\mathrm{baryon}}$ but non-negligible rest mass (e.g., a hypothetical feebly-interacting massive particle that has been in equilibrium at early times) would show a suppressed gravitational coupling proportional to its accumulated causal depth relative to baryons. This is a falsifiable prediction distinguishing RA from standard gravity.
3. A demonstration that the ρ_λ gradient correction is negligible at the scales probed by precision WEP tests (laboratory scales, $r \ll r_d$), recovering standard GR universality in the weak-field, quasi-static regime.

Collaborator target: experimental gravity, precision tests of the equivalence principle.

Operationalization of $\lambda(x)$. Prediction 1 (Hubble tension) requires an observable proxy for $\lambda(x)$. The natural candidates are:

- **Galaxy number density** $n_{\mathrm{gal}}(x)$ from large-scale structure surveys (DESI, SDSS, 2MPP). This is proportional to local stellar mass density, which is proportional to integrated star-formation history, which is correlated with long-term actualization flux. Proxy quality: high, but includes selection effects.
- **Star-formation rate density** $\dot{\rho}_*$ from infrared surveys. More directly proportional to current λ in dense regions, but misses the accumulated depth of old stellar populations.

- **Baryon density** ρ_b from X-ray surveys and Sunyaev-Zel'dovich measurements. Proportional to total mass in a region, which correlates with long-term actualization history for matter in equilibrium.

The Hubble tension prediction becomes falsifiable precisely when: (a) $\lambda(x)$ is replaced by one of these observational proxies; (b) a smoothing scale is specified (comoving $\sim$ 10–100 Mpc to match BAO-scale structure); (c) the predicted slope of H_0 residuals vs integrated density along the line of sight is computed from the RA field equation; and (d) the result is compared against existing H_0 maps from void-dominated surveys [Kenworthy et al., 2019] and cluster-dominated surveys.

Hard wall: quantitative H_0 prediction. The current prediction is qualitatively distinguishable from ΛCDM and early dark energy (which predict no line-of-sight correlation) but is not yet a numerical forecast. The minimum requirement for a testable claim is:

1. Fix $\lambda \propto \rho_b$ (the simplest proxy) and derive the predicted $\delta H_0/H_0$ as a function of the fractional overdensity δ along the line of sight.
2. Show consistency with kinetic Sunyaev-Zel'dovich (kSZ) bounds on large void models, which already constrain strong line-of-sight H_0 variations.
3. Produce a predicted slope and scatter for the H_0-density correlation using the RA field equation parameters.

Upstream dependency: the generative measure c_k. The quantitative H_0 forecast is downstream of Derivation 1 (Section 8.1, Hard Wall 1). The bandwidth-partition formula and Eridanus prediction are stated as Prediction 1b (Section 7.2). The ratio $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}} \approx 0.42$ is currently calibrated from the KBC void; it will be derived from the BDG-filter MCMC once Hard Wall 1 is resolved. **Collaborator target:** observational cosmology, large-scale structure; upstream: BDG-filter MCMC (D1).

8.7 Summary: Status of Each Derivation

Derivation 1 (Generative Algebra). *Skeleton established.* The RA-CSG extension with on-shell filter is defined; causal invariance is inherited; process algebra formulation given. Mean-field Poisson-CSG formula derived: $c_k^{\text{RA}}(x) = e^{-\mu}\mu^k/k!$ with $\mu = \lambda V_{\text{coh}} p_{\text{th}}$ (local, environment-dependent). BDG sign criterion (equation (23)) identified as the discrete RA actualization filter. Hard wall: determine V_{coh} and run BDG-filter MCMC to extract asymptotic c_k and $\bar{\lambda}_m/\bar{\lambda}_{\text{exp}}$.

Derivation 2 (Continuum Limit). *Substantially resolved at the finite-dimensional level; Lean-verified.* LLC $\to$ continuity equation proven in the $\lambda \to \infty$ limit. Metric reconstruction via Hawking-Malament theorem sketched. The Unruh reconciliation problem is resolved in finite dimensions *only*: `frame_independence` and `rindler_stationarity` are compiled theorems in `RA_AQFT_Proofs_v10.lean`, using the `Matrix.cfc_conj_unitary` lemma from Lean-QuantumInfo [Meiburg et al., 2025]. **Remaining hard wall:** full metric emergence from RA-CSG dynamics; extension of frame-independence to the infinite-dimensional type III_1 algebra setting via Araki's modular relative entropy [Araki, 1976]—specifically, proving $\Delta S_{\text{Araki}}(\rho_{\text{KMS}}\|\Omega) = 0$ under modular flow on the wedge algebra $\mathscr{A}(W)$, and that $\mathscr{P}_{\text{act}}$ commutes with the Tomita-Takesaki modular flow σ_s^{Ω}. **Collaborator target:** algebraic QFT, Tomita-Takesaki modular theory.

Derivation 3 ($\mathscr{P}_{\text{act}}$ Type-Constraint). *Substantial partial result.* Spectral definition given; vacuum energy suppression proved (Theorem 8.4); gauge covariance established (Proposition 2); DAG connection formalized. Hard walls (four): non-perturbative regime, renormalization compatibility (including treatment of Casimir/binding energy corrections), back-reaction fixed point, and—most fundamentally—Bianchi compatibility ($\nabla^\mu \langle T^{\mu\nu}_{\text{RA}} \rangle = 0$). The recommended

path forward is to reframe $\mathscr{P}_{\text{act}}$ as a covariant renormalization prescription rather than a literal projector that excludes virtual contributions from gravity.

8.8 Thermodynamic Derivation of the Actualization Tensor

While the gradient correction $\rho_\lambda = -(\xi/4\pi G)\nabla^2(\ln\lambda)$ accurately recovers flat galactic rotation curves in the non-relativistic regime (see RADM [Sandeman, 2026f]), establishing its covariant origin is paramount. Building on Jacobson's thermodynamic derivation of the Einstein equation of state [Jacobson, 1995], we map the local Rindler horizon to the RA causal severance boundary—the topological limit across which actualization directed edges have not yet formed.

The fundamental Clausius relation across this boundary is $dQ = T\,dS$. In standard general relativity the entropy variation dS is sourced entirely by the heat flux of accumulated matter, dS_A. However, because Relational Actualism defines the kinematic actualization vertex strictly as an irreversible increase in relative entanglement entropy ($\Delta S(\rho\|\sigma_0) > 0$), the background generation of the causal DAG itself constitutes a fundamental entropy flux. We therefore decompose the total entropy variation:

$$dS_{\text{total}} = dS_A + dS_\lambda\,. \tag{35}$$

To formulate dS_λ covariantly, we treat the logarithmic actualization density $\phi \equiv \ln\lambda$ as a scalar field. A constant, uniform actualization flow ($\nabla_\mu\phi = 0$) represents a steady-state vacuum, yielding no net horizon deformation. A spatial gradient in actualization density necessitates an acceleration of entropy production, requiring second derivatives of ϕ. The unique, dimensionally consistent tensor coupling this scalar to the horizon geometry is:

$$\Theta^{(\lambda)}_{\mu\nu} = \xi\Big[\nabla_\mu\nabla_\nu(\ln\lambda) - g_{\mu\nu}\Box(\ln\lambda)\Big]. \tag{36}$$

The actualization entropy flux across the horizon, generated by the null congruence k^μ, is thus $T\,dS_\lambda = \int_{\mathscr{H}} \Theta^{(\lambda)}_{\mu\nu}k^\mu k^\nu\,d\Sigma$. Because k^μ is null ($g_{\mu\nu}k^\mu k^\nu = 0$), the d'Alembertian term drops out of the surface integral, strictly mirroring the behaviour of the cosmological constant in Jacobson's original formulation.

Equating the expansion of the horizon (via the Raychaudhuri equation) with the total entropy flux yields the RA-modified field equations:

$$G_{\mu\nu} = 8\pi G\Big(T^{(A)}_{\mu\nu} + \Theta^{(\lambda)}_{\mu\nu}\Big)\,. \tag{37}$$

In the static weak-field limit, the 00-component of this tensor recovers exactly the $\nabla^2(\ln\lambda)$ Poisson modification proposed in RADM (the unified Poisson equation of [Sandeman, 2026f]).

Furthermore, by strict analogy with the Bekenstein-Hawking area-entropy relation, the coupling constant ξ is not a free parameter. It must emerge as the ratio of the discrete entropy generated per actualization vertex to the fundamental RA area element:

$$\xi \;\propto\; \frac{\langle\Delta S_{\text{vertex}}\rangle}{l^2_{\text{RA}}}. \tag{38}$$

Crucially, this connects macroscopic gravity directly to the Kinematic Coherence Bound established in RAQI [Sandeman, 2026e]. The rate limit on actualization events per coherence volume pins this entropy ratio to an observable threshold, constraining ξ via fault-tolerant quantum array decoherence limits.

Refining Hard Wall D3: The Bianchi Constraint

The contracted Bianchi identity requires $\nabla^\mu G_{\mu\nu} \equiv 0$. Applying the commutator of covariant derivatives to the scalar field $\phi = \ln\lambda$ yields:

$$\nabla^\mu \left(\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} \Box\phi\right) = R_{\mu\nu} \nabla^\mu \phi, \tag{39}$$

so that $\nabla^\mu \Theta^{(\lambda)}_{\mu\nu} = \xi R_{\mu\nu} \nabla^\mu (\ln\lambda)$. Imposing the Bianchi identity on the full modified field equations (37) yields a derived equation of motion for the actualization field:

$$\nabla^\mu T^{(A)}_{\mu\nu} + \xi\, R_{\mu\nu} \nabla^\mu (\ln\lambda) = 0. \tag{40}$$

This is not merely a gap; it is a positive result. The actualization field λ is not freely specifiable: it is dynamically constrained by the curvature it sources. Standard matter-energy satisfies $\nabla^\mu T^{(A)}_{\mu\nu} = 0$ strictly only when the actualization gradient is orthogonal to the Ricci tensor. In physically relevant configurations the two are coupled, and equation (40) governs that coupling.

Remaining hard wall. Proving that the discrete quantum measure of the causal DAG enforces equation (40) in the continuum limit—and that no anomalous momentum exchange survives the coarse-graining— is the primary open mathematical challenge of Relational Actualism cosmology. **Collaborator target:** algebraic QFT, Tomita-Takesaki modular theory, causal set theory.

Derivation 4 (Graph Cut Theorem). *Complete at the DAG level;* ***fully verified in Lean 4/Mathlib with no `sorry` tags.*** Full proof of Theorem 8.9 (Ledger Partition) with three parts; horizon corollary derived. The discrete graph-theoretic core has been verified in the Lean 4 theorem prover, including full proofs of the two sum-decomposition helper lemmas (`sum_outgoing_decompose` and `sum_incoming_decompose`). The Lean file is completely `sorry`-free for all Graph Cut results. Scope: energy bookkeeping in the discrete DAG, not unitarity. Translation to full GR boundary conditions requires Bianchi compatibility (Derivation 3, hard wall iv).

Derivation 7 (Covariant Step 4 — New, from Engine of Becoming). *Open.* The stochastic graph update in the Engine of Becoming [Sandeman, 2026e] selects one actualization vertex per algorithmic step — a description that is foliation-dependent. A covariant formulation must show that the stochastic update commutes for spacelike-separated candidates: selecting v_A then v_B produces the same graph as selecting v_B then v_A when v_A and v_B are causally disconnected. This is the Tomonaga-Schwinger analogue for the stochastic update, and it is the condition required for the full covariant quantum gravity interpretation of the algorithm. The causal invariance already established in Derivation 1 (for the deterministic graph growth) provides the nearest existing scaffolding. **Hard wall:** proving the stochastic update is causally invariant for spacelike-separated candidates, with an explicit error bound. **Collaborator target:** algebraic QFT, causal set theory.

Derivation 5 (Unified Metric Sourcing). *Relocated to RADM.* The $\nabla^2(\ln\lambda)$ conjectural mechanism, rotation curve derivation, Bullet Cluster A_{RA} argument, Tully-Fisher scaling, and CMB structure formation requirements are the primary subject of the companion paper RADM [Sandeman, 2026f]. The WIMP prohibition in Prediction 4 stands independently and is supported by the equilibrium-limit WEP recovery argument in Derivation 6.

Covariant formulation target. The weak-field correction $\nabla^2(\ln\lambda)$ is a Newtonian-limit expression. Its covariant generalisation is the d'Alembertian acting on the log-density field:

$$\Box(\ln\lambda) \;=\; g^{\mu\nu} \nabla_\mu \nabla_\nu (\ln\lambda), \tag{41}$$

where $g^{\mu\nu}$ is the emergent metric of Derivation 2. The full covariant coupling of this term to $g_{\mu\nu}$ — establishing how $\Box(\ln\lambda)$ enters the modified Einstein tensor and what its coupling constant ξ reduces to in the Newtonian limit — is a named target for the covariant field equation program in Derivation 3. Demonstrating that $\Box(\ln\lambda) \to \nabla^2(\ln\lambda)$ in the static weak-field limit is a consistency requirement; proving Bianchi compatibility of the resulting $\langle T^{\mu\nu}_{RA}\rangle + \xi\, g^{\mu\nu}\Box(\ln\lambda)$ source tensor is the associated hard wall. **Collaborator target:** general relativity, modified gravity.

Derivation 6 (WEP Recovery and λ Operationalization). *Qualitative argument established; quantitative proof and forecast outstanding.* WEP recovery for ordinary matter argued via equilibrium limit ($\rho_A \to \rho_{\text{matter}}$); WIMP non-universality reframed as $A_{\text{RA}} \approx 0$, not a violation for equilibrium species. Observable proxies for $\lambda(x)$ identified (galaxy density, SFR, baryon density). **Hard walls:** (i) formal proof of WEP recovery with explicit correction bounds; (ii) quantitative H_0-density slope from RA field equation; (iii) kSZ consistency check; (iv) fifth-force/atom-interferometry predictions for feebly-interacting species. **Collaborator targets:** precision gravity, observational cosmology, large-scale structure.

Epilogue: The Horizon of the Program — From Gravity to Gauge Symmetries

Relational Actualism strips gravity of its status as a fundamental force and reconceives it as the macroscopic thermodynamic shadow of the causal graph's accumulated topology. This suggests—if the framework is correct—that quantizing the metric is the wrong approach to quantum gravity. What requires understanding is not a quantum field on a spacetime background, but the discrete actualization events that generate the causal graph from which spacetime emerges.

In its current formulation, however, the framework is agnostic about the origins of the remaining three fundamental forces and the specific particle content of the Standard Model. RA currently borrows the Standard Model as a phenomenological engine: it uses known electromagnetic, weak, and strong cross-sections to inform the graph's transition probabilities without deriving those cross-sections from the graph structure itself. A mature theory of fundamental physics cannot borrow its interaction rules; it must derive them.

The ultimate multi-decade horizon of the RA program is the derivation of the Standard Model from the pure discrete topology of the causal DAG. What follows is a statement of the research direction, not a claim of achievement.

Particles as topological motifs. In a pure process ontology, what we call "matter" is the accumulated causal structure of on-shell actualization events. If that is so, the specific *kinds* of matter cannot be ontologically primitive; they must be structural features of the graph. The natural candidate is the *topological motif*: a localised, stable, repeating causal structure—a knot in the DAG—whose characteristic actualization frequency determines its contribution to A_{RA} and thereby its effective inertial mass. Different particle species (an electron versus an up quark) would correspond to mathematically distinct knotting patterns. This direction has precedents in the physics literature: Bilson-Thompson's braid model of preons [Bilson-Thompson, 2006], Hackett and Smolin's work on particle states in spin foam models, and Penrose's original spinor networks all explore the idea that particle quantum numbers can be read off topological invariants of discrete graph structures.

Gauge symmetries as edge-routing conservation laws. If particles are topological motifs, the forces between them must arise from the rules governing how motifs interact—how causal knots exchange edges. The continuous gauge symmetries of the Standard Model ($U(1)$, $SU(2)$, $SU(3)$) are conjectured to emerge as macroscopic approximations of discrete edge-routing and topological conservation laws on the DAG:

Electromagnetism ($U(1)$) The simplest Abelian case. Conjectured to emerge from a single discrete conservation law on directed edges—the graph-theoretic analogue of charge conservation—whose continuous limit is the $U(1)$ gauge symmetry of quantum electrodynamics.

The Weak Force ($SU(2)$) and Chirality The most structurally puzzling of the three, because the weak interaction violates parity—it couples exclusively to left-handed fermions. In graph terms, this would require the DAG to have an intrinsic handedness: a direction convention on causal edges that distinguishes left-handed from right-handed motifs. How a directed acyclic graph acquires an intrinsic chirality is an open problem with no clear current path. This is the hardest of the three gauge symmetries to derive and must be named as such.

The Strong Force ($SU(3)$) and Confinement Quarks are empirically never isolated (colour confinement). In the DAG picture, this suggests that certain topological motifs (quarks) are structurally unstable as open subgraphs: they do not achieve causal closure on their own. They must bind in specific multiplicities—triplets (baryons) or pairs (mesons)—to form a closed, stable causal subgraph. Colour confinement would then be a topological stability criterion rather than a dynamical force, an analogue of why the Markov blanket must be closed to maintain autopoiesis. The $SU(3)$ group structure would emerge from the symmetries of the allowed closure patterns.

The hard wall. These conjectures are statements of a research direction, not derivations. The hard wall is foundational: connecting the abstract topology of the causal DAG to the specific group-theoretic structure of the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ requires a mathematical bridge between discrete graph theory and continuous Lie algebra representation theory. The most promising existing frameworks for this bridge are spin foam models and loop quantum gravity (which embed $SU(2)$ in the graph structure directly) and category-theoretic approaches to topological quantum field theory. RA's contribution is to provide a clearer physical motivation for why the derivation should be attempted: if actualization events are ontologically primitive and gravity is their thermodynamic shadow, the particle content of the universe is not a brute fact but a theorem waiting to be proved.

Declarations

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Code availability: The Lean 4 formal proof files (`RA_AQFT_Proofs_v10.lean` and `RA_Proofs_Lean4.lean`), Python computational scripts, and all companion paper source files are publicly available at github.com/jsandeman/Relational-Actualism.

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